CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Second Semester of M. Sc. (Mathematics) Examination May 2018

MA721 : Algebra - 1

May 07, 2018, Monday | Time 10.00 a.m. to 10.45 p.m. | Maximum marks 20

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. In this question paper, $\mathbb{N}$ denotes the set of all natural numbers. Other notations are standard.

Multiple Choice Questions (MA721 : Algebra - 1)

Q-I Choose the correct answer from the given options in the following: [20]

(1) Let a, b, c $\in \mathbb{N}$ such that gcd(a, b) = 1. If $a \mid bc$, then ___________.

(a) $a \mid c$ (b) $a \mid b$ (c) $c \mid a$ (d) $\mathrm{lcm}(b, c) \mid a$

(2) Let a, b, c $\in \mathbb{N}$ and d = gcd(a, b). Then the Diophantine equation ax + by = c has an integral solution if ______________.

(a) $d \mid a$ (b) $c \mid d$ (c) $b \mid a$ (d) $d \mid c$

(3) Let φ be an Euler function; p be a prime number and k $\in \mathbb{N}$. Then $\varphi(p^k)$ = _____.

(a) p (b) pk (c) $p^k - p^{k-1}$ (d) $p^k\, p^{k-1}$

(4) Let G be a finite group and H be a normal subgroup of G. Then o(G/H) = ____.

(a) o(G) / o(H) (b) o(G) o(H) (c) o(G) + o(H) (d) o(G) – o(H)

(5) Every group of order four is ________.

(a) simple (b) abelian (c) cyclic (d) nonabelian

(6) Let H and K be subgroups of a finite group G such that o(H)o(K) = o(G) and $H \cap K = \{e\}$. Then _____________

(a) HK need not be a subgroup (c) HK is a proper subgroup of G

(b) HK = G (d) neither HK = G nor HK is a subgroup

(7) Let G be a group of nonzero real numbers under usual multiplication. Define $\varphi: G \to G$ by _____________, $x \in G$. Then φ is a homomorphism.

(a) $\varphi(x) = x + 5$ (b) $\varphi(x) = x - 5$ (c) $\varphi(x) = 5^x$ (d) $\varphi(x) = x^5$

(8) Let A_n be an alternating group of n objects. Then $o(A_n)$ = _______.

(a) n (b) n! (c) n!/2 (d) n! + 2

(9) Let G be an infinite cyclic group. Then the group of all automorphisms on G, is isomorphic to _______.

(a) a cyclic group of order $\neq 2$ (b) a cyclic group of order 2 (c) G (d) none of these

(10) Let G be a group of order 36 and H be a subgroup of G of order 9. Then ________.

(a) H is normal (b) H is not abelian

(c) H contains a nontrivial normal subgroup (d) none of these

(11) Let G be a group. For $g \in G$, define $\lambda_g : G \to G$ by $\lambda_g(x) = xg$, $x \in G$. Then

(a) λ_g is a homomorphism (b) $\lambda_{gh} = \lambda_{hg}$ for all g, h $\in$ G.

(c) $\lambda_{gh} = \lambda_g \lambda_h$ for all g, h $\in$ G (d) $\lambda_{gh} = \lambda_h \lambda_g$ for all g, h $\in$ G.

(12) Let G be an abelian group. Then the number of inner automorphisms of G is __.

(a) 1 (b) o(G) (c) o(G)/2 (d) 0

(13) The product of two disjoint three cycles is ________

(a) odd (b) even (c) a three cycle (d) none of these

(14) Let G be a group and Z(G) be its center. If $a \in Z(G)$, then c_a = ______, where c_a is the number of elements in G conjugate to a.

(a) 1 (b) 0 (c) o(G) (d) none of these

(15) Let G be a group of order 125. Then ________.

(a) o(Z(G)) = 0 (b) o(Z(G)) = 1 (c) o(Z(G)) $\geq$ 5 (d) o(Z(G)) = 15

(16) A finite group of prime order is __________.

(a) nonabelian (b) simple (c) not simple (d) not soluble

(17) Let F be a field of characteristic 5 and GL(2, F) be the group of all 2×2 regular matrices over F. The o(GL(2, F)) = ________.

(a) 7 (b) 10 (c) 240 (d) 480

(18) A group of order 51 is _______.

(a) cyclic (b) simple (c) not abelian (d) none of these

(19) Let G be a group of order 64. Then G is ________.

(a) abelian (b) simple (c) soluble (d) not soluble

(20) Let G be a group of order 20. Then the number of 5-Sylow subgroups of G is _____.

(a) 4 (b) 2 (c) 5 (d) 1

Candidate's seat number__________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of M. Sc. (Mathematics) Examination May 2018

MA721 : Algebra - 1

May 07, 2018, Monday   Time 10.45 a.m. to 01.00 p.m.   Maximum marks 50

Instructions:

1. Section I and Section II must be written in separate answer books.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. In this question paper, $\mathbb{N}$ denotes the set of all natural numbers. Other notations are standard.

SECTION-I

Q-II(A) Attempt any TWO of the following: [8]

(1) State and prove Chines Remainder theorem.

(2) Let X be a set; G be a group act on X and $x \in X$. Define orbit of x. Show that the number of elements in orbit of $x = \frac{o(G)}{o(G_x)}$, where $G_x = \{ g \in G : g \cdot x = x \}$ is the stabilizer of x.

(3) Show that a group of order p^2 is abelian, where p is a prime number.

(B) Attempt any SIX of the following: [12]

(1) Let a, b be non-zero integers. Show that a and b are relatively prime iff there are integers x and y such that $ax + by = 1$.

(2) Let H and K be subgroups of a group G such that HK = KH. Show that HK is a subgroup of G.

(3) Show that the quaternion group Q_8 is soluble

(4) Let G be a group and H be a nonempty finite subset of G such that H is closed under multiplication. Show that H is a subgroup of G.

(5) Let G be a group and Z(G) be its center. If G/Z(G) is cyclic, show that G is abelian.

(6) Find orbits of 1 and 2 under the permutation $\theta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 1 & 3 & 2 \end{pmatrix}$. Represent θ as a product of transpositions.

(7) Let X be a set; G be a group act on X and $x \in X$. Show that the stabilizer $G_x = \{ g \in G : g \cdot x = x \}$ of x is a subgroup of G.

(8) Let G be group; H be a normal subgroup of G and K be a characteristic subgroup of K. Show that K is a normal subgroup of G.

(9) Let G be a group and H be a normal subgroup of G. Show that G/H is abelian iff $G' \subset H$, where G' is a commutator subgroup of G. Hence show that G/G' is abelian.

SECTION-II

Q-III(A) Attempt any THREE of the following: [18]

(1) Let G be a finite abelian group and p be a prime number such that p | o(G). Show that there is $e \neq a \in G$ such that $a^p = e$.

(2) Let G be a finite group and $a \in G$. Show that $c_a = \frac{o(G)}{o(N(a))}$, where c_a is the number of elements of G conjugate to a.

(3) Let G be a finite group and p be a prime number such that p divides o(G). Show that there is p-Sylow subgroup of G.

(4) Let G be a finite abelian group of order $n = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_k^{\alpha_k}$, where $p_1, p_2, \ldots. p_k$ are distinct primes and $\alpha_j \in \mathbb{N}$ for j = 1,2,3,…,k.

Let P_j be the p_j-Sylow subgroup of G for j = 1,2,3,…,k. Show that G is the internal direct product of $P_1, P_2, \ldots., P_k$.

(5) Let G, $\overline{G}$ be groups and $\varphi : G \to \overline{G}$ be an onto homomorphism with kernel K. Let $\overline{N}$ be a normal subgroup of $\overline{G}$. Define N = { $g \in G: \varphi(g) \in \overline{N}$ }. Show that G/N is isomorphic to $\overline{G} / \overline{N}$.

(B) Attempt any THREE of the following: [12]

(1) Let G be a group; H be a subgroup of G and $a \in G$. Show that
Ha = $\{x \in G : a \equiv x \bmod H\}$.

(2) Let G be a group. Show that there is a set S such that G is isomorphic to some subgroup of A(S).

(3) Show that the number of conjugate classes of the symmetric group S_n is p(n), the number of partitions of n.

(4) Let G be a group and N be a normal subgroup of G. Show that G is soluble iff G/N and N both are soluble.

(5) Let G be a group of order p^k, where p is a prime number and $k \in \mathbb{N}$. Show that Z(G), the center of G, is non trivial.

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